

MAT 528 HW01

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Problem 1 (Munkres 13.4). (a) If $\{\mathcal{T}_\alpha\}$ is a family of topologies on X , show that $\bigcap \mathcal{T}_\alpha$ is a topology on X . Is $\bigcup \mathcal{T}_\alpha$ a topology on X ?

(b) Let $\{\mathcal{T}_\alpha\}$ be a family of topologies on X . Show that there is a unique smallest topology on X containing all the collections $\{\mathcal{T}_\alpha\}$, and a unique largest topology contained in all the $\mathcal{T}_\alpha$.

(c) If $X = \{a, b, c\}$, let

$$\mathcal{T}_1 = \{\emptyset, X, \{a\}, \{a, b\}\} \quad \text{and} \quad \mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b, c\}\}.$$

Find the smallest topology containing $\mathcal{T}_1$ and $\mathcal{T}_2$, and the largest topology contained in $\mathcal{T}_1$ and $\mathcal{T}_2$.

Solution. (a) We check that $\bigcap_\alpha \mathcal{T}_\alpha$ satisfies the three axioms for a collection of subsets on X to be a topology.

- Each of the $\mathcal{T}_\alpha$ contains X and the empty set, so the intersection of the $\mathcal{T}_\alpha$ also contains X and $\emptyset$.
- Suppose that $\{U_\beta\}_{\beta \in B}$ is a collection of sets in $\bigcap_\alpha \mathcal{T}_\alpha$. Then $\{U_\beta\}_{\beta \in B} \subseteq \mathcal{T}_\alpha$ for each $\alpha \in A$, so $\bigcup_{\beta \in B} U_\beta$ is in each $\mathcal{T}_\alpha$, hence $\bigcup_{\beta \in B} U_\beta$ is in the intersection of the $\mathcal{T}_\alpha$.
- Finally, suppose that $U_1, \dots, U_n$ are n sets in the intersection of the $\mathcal{T}_\alpha$. Then these n sets are contained in $\mathcal{T}_\alpha$ for all $\alpha \in A$, so their intersection is in each $\mathcal{T}_\alpha$, hence $U_1 \cap \dots \cap U_n$ lies in $\bigcap_\alpha \mathcal{T}_\alpha$.

Therefore the intersection of the $\mathcal{T}_\alpha$ is a topology on X .

On the other hand, unions of topologies need not be topologies. For example, $\mathcal{T}_1 = \{\emptyset, X, \{a\}\}$ and $\mathcal{T}_2 = \{\emptyset, X, \{b\}\}$ are topologies on $X = \{a, b, c\}$, but their union $\mathcal{T}_1 \cup \mathcal{T}_2$ is not a topology because it contains $\{a\}$ and $\{b\}$ but not their union $\{a, b\}$.

(b) Let $\mathcal{T}_L$ be the intersection of all topologies on X containing $\bigcup_\alpha \mathcal{T}_\alpha$. Clearly this is a topology containing all the $\mathcal{T}_\alpha$ that is minimal with respect to inclusion, since if $\mathcal{T}'_L$ is a topology containing all the $\mathcal{T}_\alpha$ then it is part of the intersection that generated $\mathcal{T}_L$, hence $\mathcal{T}_L \subseteq \mathcal{T}'_L$. This topology is also the *unique* minimal topology containing all the $\mathcal{T}_\alpha$ because if $\mathcal{T}''_L$ is another minimal topology then by minimality $\mathcal{T}_L$ and $\mathcal{T}''_L$ contain each other.

Similarly, the largest topology contained in all the $\mathcal{T}_\alpha$ is $\bigcap_\alpha \mathcal{T}_\alpha$. This is a topology by part (a) and it is easy to see that it is the unique largest one because $\bigcap_\alpha \mathcal{T}_\alpha$ is straight up the largest set contained in all of the $\mathcal{T}_\alpha$.

(c) The smallest topology containing both is

$$\{\emptyset, X, \{a\}, \{b\}, \{a, b\}, \{b, c\}\}$$

since it must contain $\mathcal{T}_1 \cup \mathcal{T}_2$ of course, and in addition $\{b\} = \{a, b\} \cap \{a, c\}$ is in this topology as well, which accounts for all 6 sets. The largest topology contained in $\mathcal{T}_1$ and $\mathcal{T}_2$ is just their intersection

$$\{\emptyset, X, \{a\}\}. \quad \square$$

Problem 2 (Munkres 13.5). Show that if $\mathcal{A}$ is a basis for a topology on X , then the topology generated by $\mathcal{A}$ equals the intersections of all topologies on X that contain $\mathcal{A}$. Prove the same if $\mathcal{A}$ is a subbasis.

Solution. By the previous problem, the topology generated by $\mathcal{A}$ must contain the intersection of all topologies containing $\mathcal{A}$ because the intersection is minimal with respect to inclusion among topologies containing $\mathcal{A}$. Conversely, any set in the topology $\mathcal{T}$ generated by $\mathcal{A}$ is the union of some sets in $\mathcal{A}$. Since topologies are closed under arbitrary unions, every topology containing $\mathcal{A}$ contains arbitrary unions of elements of $\mathcal{A}$, so all sets in $\mathcal{T}$ are contained in every topology containing $\mathcal{A}$, which means the topology generated by $\mathcal{A}$ is contained by the intersection of all topologies containing $\mathcal{A}$.

If $\mathcal{A}$ is a subbasis then again the topology generated by $\mathcal{A}$ contains the intersection of all topologies containing $\mathcal{A}$. For the converse, we note that open sets in the topology generated by $\mathcal{A}$ are arbitrary unions of finite intersections of sets in $\mathcal{A}$. Topologies are closed under finite intersections and arbitrary unions, so any topology containing $\mathcal{A}$ contains all finite intersections in $\mathcal{A}$ and thus all arbitrary unions of finite intersections of sets in $\mathcal{A}$. That is, every topology on X containing $\mathcal{A}$ contains the topology generated by $\mathcal{A}$, i.e. the intersection of all topologies on X containing $\mathcal{A}$ contains the topology generated by $\mathcal{A}$. $\square$

Problem 3 (Munkres 13.8a). Apply Lemma 13.2 to show that the countable collection

$$\mathcal{B} = \{(a, b) : a < b, a \text{ and } b \text{ are rational}\}$$

is a basis that generates the standard topology on $\mathbb{R}$.

Solution. Let x be an element of $\mathbb{R}$ and $U \subseteq \mathbb{R}$ an open set containing x . Then, U contains some interval $(x - \varepsilon, x + \varepsilon)$ centered at x , since the standard topology on $\mathbb{R}$ is defined with open intervals as a basis. Pick rational numbers $a \in (x - \varepsilon, x)$ and $b \in (x, x + \varepsilon)$; then $x \in (a, b) \subseteq U$ and $(a, b) \in \mathcal{B}$. By Lemma 13.2 this is enough to imply that $\mathcal{B}$ is a basis for the standard topology on x . $\square$

Problem 4 (Munkres 16.3). Consider the set $Y = [-1, 1]$ as a subspace of $\mathbb{R}$. Which of the following sets are open in Y ? Which are open in $\mathbb{R}$?

$$\begin{aligned} A &= \{x : \tfrac{1}{2} < |x| < 1\} \\ B &= \{x : \tfrac{1}{2} < |x| \leq 1\} \\ C &= \{x : \tfrac{1}{2} \leq |x| < 1\} \\ D &= \{x : \tfrac{1}{2} \leq |x| \leq 1\} \\ E &= \{x : 0 < |x| < 1 \text{ and } 1/x \notin \mathbb{Z}_+\} \end{aligned}$$

Solution. • A is open in $\mathbb{R}$ because it is the union of intervals $(-1, -\frac{1}{2}) \cup (\frac{1}{2}, 1)$, and it is open in Y because $A = A \cap Y$ where A is open in $\mathbb{R}$.

- B is not open in $\mathbb{R}$ because any interval centered at 1 must contain some points greater than 1, which means said interval is not inside B . On the other hand, B is open in Y because $B = ((-2, \frac{1}{2}) \cup (\frac{1}{2}, 2)) \cap Y$.

- C is not open in $\mathbb{R}$ because any interval centered at $\frac{1}{2}$ contains points outside C (specifically some points just below $\frac{1}{2}$). It is also not open in Y because of the same reason, since if $C = U \cap Y$ for some $U \subseteq \mathbb{R}$ then U would have the same behavior around $\frac{1}{2}$, so U isn't open in $\mathbb{R}$.
- D is not open in $\mathbb{R}$ again because any interval around $\frac{1}{2}$ must contain some points greater than 1 so the interval isn't contained within D . It is also again not open in Y because if $D = U \cap Y$ for some $U \subseteq \mathbb{R}$ then U cannot be open in $\mathbb{R}$ because it has the same behavior around $\frac{1}{2}$.
- E is open in $\mathbb{R}$ because it is the union of open intervals

$$\bigcup_{n=1}^{\infty} \left(\frac{1}{n+1}, \frac{1}{n} \right) \cup \bigcup_{n=1}^{\infty} \left(-\frac{1}{n}, -\frac{1}{n+1} \right)$$

and it is open in Y because each of the intervals in this union is strictly contained within Y , hence also open in Y . □

Problem 5 (Munkres 16.4). A map $f: X \rightarrow Y$ is said to be an **open map** if for every open set U of X , the set $f(U)$ is open in Y . Show that $\pi_1: X \times Y \rightarrow X$ and $\pi_2: X \times Y \rightarrow Y$ are open maps.

Solution. Recall that a basis for the product topology on X is the collection of sets $\{U_X \times U_Y : U_X \subseteq X, U_Y \subseteq Y\}$. If $U \subseteq X \times Y$ is open, then for each $p \in U$ there is a basis element $U_{X,p} \times U_{Y,p}$ containing p with $U_{X,p} \subseteq X$ and $U_{Y,p} \subseteq Y$. When projected onto X , we get

$$\pi_1(U) = \pi_1 \left(\bigcup_{p \in U} U_{X,p} \times U_{Y,p} \right) = \bigcup_{p \in U} \pi_1(U_{X,p} \times U_{Y,p}) = \bigcup_{p \in U} U_{X,p} \subseteq X$$

which is the union of open subsets of X , hence is open in X . Similarly

$$\pi_2(U) = \bigcup_{p \in U} U_{Y,p} \subseteq Y$$

is open in Y because it is the union of open subsets of Y . □

Problem 6 (Munkres 16.6). Show that the countable collection

$$\{(a, b) \times (c, d) : a < b \text{ and } c < d \text{ and } a, b, c, d \text{ are rational}\}$$

is a basis for $\mathbb{R}^2$.

Solution. This is an immediate corollary of the fact that if $\mathcal{B}_X$ and $\mathcal{B}_Y$ are bases for topologies on X and Y , then

$$\mathcal{B}_{X \times Y} = \{U \times V : U \in \mathcal{B}_X, V \in \mathcal{B}_Y\}$$

is a basis for the product topology on $X \times Y$ — in this case, $X = Y = \mathbb{R}^2$ and $\mathcal{B}_X = \mathcal{B}_Y$ is the basis consisting of intervals with rational endpoints, as per the result of Problem 3 (Munkres 13.8a). □

Problem 7 (Munkres 16.8). If L is a straight line in the plane, describe the topology L inherits as a subspace of $\mathbb{R}_\ell \times \mathbb{R}$ and as a subspace of $\mathbb{R}_\ell \times \mathbb{R}_\ell$.

Solution. The basis elements of $\mathbb{R}_\ell \times \mathbb{R}$ are of the form $[a, b) \times (c, d)$ for reals $a < b$, $c < d$; intersecting them with L gives a basis for the subspace topology on L . If L is vertical, then L is homeomorphic to $\mathbb{R}$ with the standard topology because the intersection of a basis set with L is either empty or a set of the form $\{x_0\} \times (c, d)$ where x_0 is the x -coordinate of the points on the line. If L isn't vertical, then L is homeomorphic to $\mathbb{R}_\ell$ because the segment along which L intersects a basis set either has its left endpoint or neither endpoint, and the latter case covered by the former because $\mathbb{R}_\ell$ is finer than $\mathbb{R}$ with the standard topology.

The basis elements of $\mathbb{R}_\ell \times \mathbb{R}_\ell$ are of the form $[a, b) \times [c, d)$ for real $a < b$, $c < d$. If L has negative slope then for any point $(x, y) \in L$,

$$L \cap [x, x+1) \times [y, y+1) = (x, y)$$

is open in L , so L just gets the discrete topology. Otherwise, L is homeomorphic to $\mathbb{R}_\ell$ because every intersection of L with a basis set will be a segment with its left endpoint but not its right endpoint. $\square$